Fire & Gas Detection Layout - Preliminary Design Report

Document DEMO-FG-RPT-001 · Revision A (issued for review) · 27 Oct 2026 · Synthetic demonstration data

1. Scope and basis of design

This report describes a preliminary fire and gas (F&G) detector layout for a synthetic gas compression station. The layout was produced automatically from the plot plan by DetectorPlace and is intended as a starting point for review by a competent F&G engineer. It covers 1827 m² of graded fire hazard area and 657 m² of credible gas release zones around 8 leak sources.

2. Method

Equipment footprints, elevation bands, hazard grades and leak sources were extracted from the plot-plan drawing by computer vision (equipment IoU 99.3% against the source geometry). 4,432 candidate flame detector configurations (position, mounting height and orientation) and 233 gas detector positions were evaluated by 3D line-of-sight ray marching (8,250,345 detector-to-point checks). A multi-objective genetic algorithm (NSGA-II with repair and redundancy pruning) traded installed cost against risk-weighted coverage shortfall. The cheapest layout meeting every target was selected.

3. Performance targets and achieved coverage

Targets follow the performance-based structure of ISA-TR84.00.07: a minimum coverage fraction per hazard grade, with voted (2ooN) coverage required in Grade A areas to support executive action. Target values are project assumptions and must be confirmed against the site F&G philosophy.

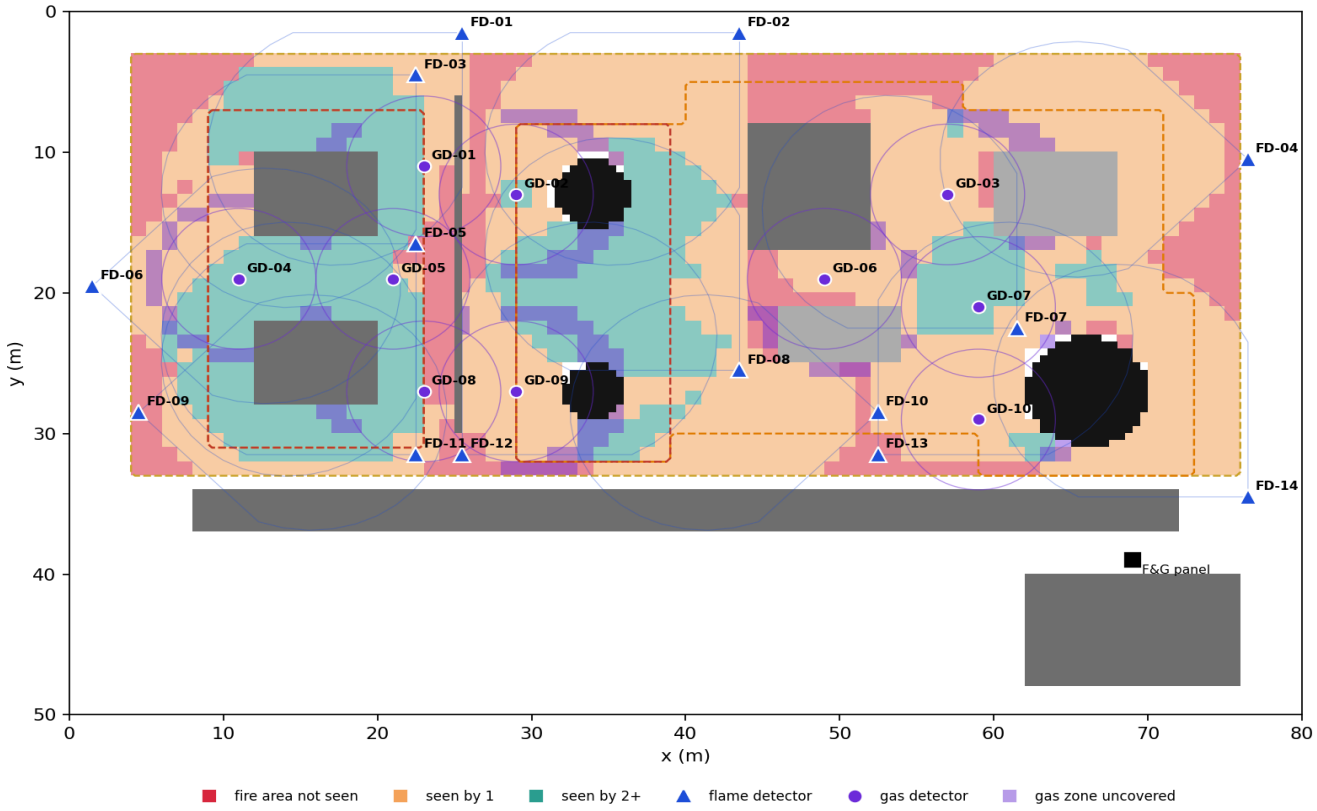
Requirement	Target	Achieved	Status
Grade A fire, 1ooN	90%	99.1%	Met
Grade A fire, 2ooN	75%	77.9%	Met
Grade B fire, 1ooN	85%	88.6%	Met
Grade C fire, 1ooN	60%	66.2%	Met
Gas release zones, 1ooN	80%	82.3%	Met

4. Detector schedule

14 flame detectors and 10 point gas detectors are proposed. Coordinates are in metres from the plot-plan origin (top-left); azimuth is measured clockwise from plant east on the drawing.

Tag	Type	x (m)	y (m)	Mount (m)	Azimuth	Unique m²	Cost (£)
FD-01	Flame (IR3)	25.5	1.5	3.0	135°	46	7,148
FD-02	Flame (IR3)	43.5	1.5	3.0	135°	132	6,644
FD-03	Flame (IR3)	22.5	4.5	3.0	135°	13	7,148
FD-04	Flame (IR3)	76.5	10.5	3.0	180°	143	5,888
FD-05	Flame (IR3)	22.5	16.5	3.0	135°	3	6,812
FD-06	Flame (IR3)	1.5	19.5	3.0	0°	56	7,316
FD-07	Flame (IR3)	61.5	22.5	3.0	225°	101	5,552
FD-08	Flame (IR3)	43.5	25.5	3.0	225°	47	5,972
FD-09	Flame (IR3)	4.5	28.5	3.0	0°	39	6,980
FD-10	Flame (IR3)	52.5	28.5	3.0	180°	106	5,636
FD-11	Flame (IR3)	22.5	31.5	3.0	225°	0	6,392
FD-12	Flame (IR3)	25.5	31.5	3.0	315°	71	6,308
FD-13	Flame (IR3)	52.5	31.5	3.0	315°	114	5,552
FD-14	Flame (IR3)	76.5	34.5	3.0	225°	109	5,216
GD-01	Point gas (IR)	23.0	11.0	0.5	-	39	4,752
GD-02	Point gas (IR)	29.0	13.0	0.5	-	66	4,528
GD-03	Point gas (IR)	57.0	13.0	0.5	-	56	3,744
GD-04	Point gas (IR)	11.0	19.0	0.5	-	72	4,864
GD-05	Point gas (IR)	21.0	19.0	0.5	-	46	4,584
GD-06	Point gas (IR)	49.0	19.0	0.5	-	39	3,800
GD-07	Point gas (IR)	59.0	21.0	0.5	-	37	3,464
GD-08	Point gas (IR)	23.0	27.0	0.5	-	39	4,304
GD-09	Point gas (IR)	29.0	27.0	0.5	-	68	4,136
GD-10	Point gas (IR)	59.0	29.0	0.5	-	49	3,240

5. Coverage map



6. Single-detector failure (N-1) check

Losing any one of 24 detectors (FD-01, FD-02, FD-03, FD-04, FD-05, FD-06, FD-07, FD-08...) drops at least one requirement below target. This is normal for a cost-minimal layout; the project should decide whether N-1 compliance is required for Grade A areas, which would add detectors.

Out of service	A1	A2	B1	C1	G1	Below target
FD-01	98.9	63.8	88.6	60.2	82.3	A2
FD-02	93.3	72.9	83.6	56.2	82.3	A2, B1, C1
FD-03	99.1	63.8	88.6	64.4	82.3	A2
FD-04	99.1	77.9	78.5	55.5	82.3	B1, C1
FD-05	98.4	60.2	88.6	66.2	82.3	A2
FD-06	95.3	62.0	88.6	61.0	82.3	A2
FD-07	99.1	77.9	72.5	66.2	82.3	B1
FD-08	97.8	53.7	83.1	65.2	82.3	A2, B1
FD-09	98.4	66.7	88.6	61.4	82.3	A2
FD-10	96.6	70.0	79.5	61.1	82.3	A2, B1
FD-11	99.1	64.9	88.6	66.2	82.3	A2
FD-12	92.8	58.8	88.6	60.5	82.3	A2
FD-13	99.1	77.9	72.0	64.8	82.3	B1
FD-14	99.1	77.9	77.3	61.1	82.3	B1
GD-01	99.1	77.9	88.6	66.2	76.4	G1
GD-02	99.1	77.9	88.6	66.2	72.3	G1
GD-03	99.1	77.9	88.6	66.2	73.8	G1
GD-04	99.1	77.9	88.6	66.2	71.4	G1
GD-05	99.1	77.9	88.6	66.2	75.3	G1
GD-06	99.1	77.9	88.6	66.2	76.4	G1
GD-07	99.1	77.9	88.6	66.2	76.7	G1
GD-08	99.1	77.9	88.6	66.2	76.4	G1
GD-09	99.1	77.9	88.6	66.2	72.0	G1
GD-10	99.1	77.9	88.6	66.2	74.9	G1

7. Residual uncovered areas

5 uncovered regions of 4 m² or more remain in Grade A/B areas (largest 31 m²). Each is listed with the reason it is uncovered.

Location (x, y)	Grade	Area m²	Reason
(49.8, 6.3)	B	31	Visible from available positions but left uncovered because the grade target is already met; covering it would cost an extra detector.
(45.9, 20.6)	B	21	Visible from available positions but left uncovered because the grade target is already met; covering it would cost an extra detector.
(51.3, 27.2)	B	6	Visible from available positions but left uncovered because the grade target is already met; covering it would cost an extra detector.
(59.5, 12.0)	B	4	Visible from available positions but left uncovered because the grade target is already met; covering it would cost an extra detector.
(61.0, 32.5)	B	4	Visible from available positions but left uncovered because the grade target is already met; covering it would cost an extra detector.

8. Cost estimate (Class 4, indicative)

Rates are placeholders and must be replaced with current supplier quotations and company estimating norms before use.

Item	Qty	Amount (£)
Flame detectors, supplied and installed	14	60,200
Upper-level mounting brackets / access	0	0
Point gas detectors, supplied and installed	10	21,000
Field cabling to F&G panel (routed)	-	41,580
Panel I/O, termination, commissioning	24	7,200
Panel, engineering, cause & effect, FAT (fixed)	1	18,000
Total		147,980

9. Limitations and verification

Geometric mapping assumes a 2.5D plant (equipment as extruded footprints), a fixed design fire height and an idealised detector cone with off-axis range loss. It does not model congestion below pipe racks, smoke obscuration, gas dispersion (CFD) or detector-specific polar diagrams. Results were re-checked against the source geometry (A1 98.7%, A2 77.2%, B1 87.3%, C1 68.1%, G1 82.6%). This layout is advisory. It must be verified and signed off by a competent person before construction.

Sign-off

Prepared (DetectorPlace v0.1, automated)	Name:	Date:
Checked (F&G engineer)	Name:	Date:
Approved (lead engineer)	Name:	Date: